

AI/ML Project Ideas with Research Papers

Guys, I am sharing a few more AI/ML project ideas along with research papers, paper links, and short project descriptions. We can discuss these ideas together and select the best one for our final proposal.

1. AI-Based Crowd Counting System

Project Idea: Estimate crowd density and count people from CCTV footage using deep learning and computer vision techniques.

Features:

- Crowd counting
- Heatmap generation
- Alert system for overcrowding
- Real-time analytics dashboard

Technologies: CNN, YOLO, Density Maps, OpenCV

Research Papers & Links:

- **CSRNet – “CSRNet: Dilated Convolutional Neural Networks for Crowd Counting”**
<https://arxiv.org/abs/1802.10062>
 - **ShanghaiTech Dataset – “Large-Scale Crowd Counting Dataset”**
<https://paperswithcode.com/dataset/shanghaitech>
-

2. AI-Based Handwritten Text Recognition

Project Idea: Convert handwritten notes or documents into editable digital text using OCR and deep learning models.

Features:

- OCR from images
- Multi-language support
- PDF export
- Real-time handwriting recognition

Technologies: CNN, LSTM, OCR, CRNN

Research Papers & Links:

- **CRNN – “An End-to-End Trainable Neural Network for Image-Based Sequence Recognition”**
<https://arxiv.org/abs/1507.05717>
 - **IAM Handwriting Database**
<https://fki.tic.heia-fr.ch/databases/iam-handwriting-database>
-

3. Sign Language Recognition System

Project Idea: Convert sign language gestures into text or speech using computer vision and deep learning.

Features:

- Real-time webcam detection
- Text generation
- Voice output
- Alphabet and gesture recognition

Technologies: OpenCV, MediaPipe, CNN, LSTM

Research Papers & Links:

- **MediaPipe Hands – “Real-Time Hand Tracking with MediaPipe”**

<https://arxiv.org/abs/2006.10214>

- **Sign Language Recognition using Deep Learning**

<https://ieeexplore.ieee.org/search/searchresult.jsp?queryText=sign+language+recognition+deep+learning>

If you have more project ideas, please share them as well so we can discuss and consider them for our proposal.

Thank you!